

An approach for accurate identification and monitoring of species in mangrove forests based on multi-source spectral data and deep learning

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ABSTRACT

The mangroves conservation is essential due to the wide range of goods and ecosystem services. Despite their importance, mangrove distribution has drastically decreased in recent decades, highlighting the need to enhanced conservation efforts through innovative monitoring methods. Accordingly, this study presents an approach that integrates remote sensing data in a study area with a diverse range of ecological scenarios, comprising monospecific mangrove forests, which are dominated by a single species and mixed mangroves with flooded freshwater forested wetlands. First, the integration of photogrammetric analysis from unmanned aerial vehicles (UAVs) enables the precise determination of the spatial distribution of mangrove forests. Furthermore, the multispectral data obtained from the Sentinel-2 satellite was employed for the training of multilayer perceptron models (MLP) that are capable of accurately mapping mixed and monospecific mangrove forests. The computational experiments for MLP models involve different number of neurons and hidden layers. The best models for detecting mangrove forest achieved an accuracy of 99.95 % for training and 99.8 % for test, while for monospecific mangrove forest, at the species level, it attained an accuracy of 98.73 % for training and 96.84 % for test, in both cases the models demonstrated a great performance test for segmentation of different kind of mangrove forests. By leveraging multi-source data and the capabilities of remote sensing and artificial intelligence technologies, this approach offers a groundbreaking solution for monitoring mangrove ecosystems. Also, this minimizes the economic and human efforts and reduces risk of error, thus leading to more efficient and effective ecosystem conservation.

1. Introduction

Mangroves are evergreen forests that are found in tropical and subtropical regions. The composition of mangrove forests can vary considerably, contingent on the presence, abundance, diversity and dominance of species. They may be classified as monospecific forests when a single mangrove species is predominant, or as mixed forests when multiple species are present, in conjunction with other wetland vegetation. In general, they can tolerate high salt levels and grow on soft

substrates along the land-sea interface in intertidal environments such as coastlines, shallow lagoons, estuaries, rivers, and deltas. These ecosystems cover approximately 14.8 million hectares across 124 countries in Asia, America, Africa, and Oceania. Recent estimation suggests the existence of 50 to more than 70 mangrove species (FAO, 2007, 2023). Mangroves provide a variety of ecosystem services, including regulating, supporting, provisioning and cultural services. Regulating ecosystem services maintain essential ecological processes and are the life support system; supporting provide adequate space for animal and plant species

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to live and develop, thus providing shelter and protection for the gene pool; provisioning refers to food for humans and animals, resource supply, raw materials, mineral, timber, medicinal, genetic and ornamental resources; and cultural refers to aesthetics, spirituality, cultural inspiration and art, as well as knowledge generation for cognitive development in education and science. They thus provide direct and indirect human benefits that can be valued at up to US \$194,000 per hectare per year (Costanza et al., 2014; Moreno-Casasola, 2016).

It is recognized that there has been a loss of mangrove forests worldwide. Since the beginning of the 21st century, approximately 4.3 % of mangroves have been lost as a result of land use change for livestock, aquaculture, urbanization and tourism development (Moreno Casasola et al., 2002; Trouche et al., 2016). Consequently, numerous studies have concentrated on monitoring changes in mangrove forests at local, regional, and global scales since 1980 (CONABIO, 2016, 2021; FAO, 2023; López-Portillo and Ezcurra, 2002; Rodríguez-Zuñiga et al., 2013; Siikamäki et al., 2012; Velázquez and Luna, 2007). The Food and Agriculture Organization (FAO) and the United Nations Environment Programme (UNEP) in collaboration with mangrove specialists worldwide, have estimated 15.2 million hectares in 2005, compared to an estimated total global mangrove area of 16.9 million hectares in 1990 (FAO, 2007). Various methods have been used to determine the distribution of mangroves, such as aerial photography, field patrols, and satellite sensors such as Sentinel-2, Landsat, Pléiades, WorldView 2, Spot, and Rapid Eye (Chopade et al., 2023; Rahmandhana et al., 2022; Trouche et al., 2016; Wang et al., 2018).

Remote sensing has enabled the monitoring of mangroves at various scales by constantly gathering information from large areas and various electromagnetic spectrum ranges. According to the United Nations Office for Outer Space Affairs (UNOOSA), over 1000 earth observation satellites were operational in 2023 with different spatial, spectral, and temporal resolutions (Chuvieco, 2009; FAO, 2023; Vázquez-González et al., 2017).

Although a significant proportion of this data is available at commercial prices, there are few free options. In the case of Mexico, a continuous monitoring program has been conducted every five years since 2005, utilizing the Moderate Resolution Imaging Spectroradiometer (MODIS) satellite system with a spatial resolution of 250 m to identify changes in mangrove vegetation and estimation for biomass (Rodríguez Zúñiga et al., 2018). Also, SPOT satellite imagery with a resolution of 10 m in the visible and infrared range were employed for the period from 2003 and stopped in 2007, due to the economic implications of doing so (CONABIO, 2009; Velázquez-Salazar et al., 2021). This leads to considerable financial and human resource costs associated with monitoring, coupled with the inconsistency of multispectral data and spatial resolution when using different satellites that cannot be compared. However, some satellites, such as Landsat and Sentinel-2, provide free access and are often used in vegetation monitoring studies. Landsat has been providing data since the 1980s with a 16-day periodicity, 30-m spatial resolution and 10 bands at different wavelengths. Sentinel-2, launched almost a decade ago, provides images with a 10-day periodicity, a spatial resolution of up to 10 m and 12 bands at different wavelengths (Chopade et al., 2023; Cissell et al., 2017; Rahmandhana et al., 2022).

Nevertheless, new technologies such as artificial intelligence can now be used to improve monitoring methods (Aviña-Hernández et al., 2023; Jhonnerie et al., 2015). The fusion of multi-source data, including artificial intelligence and remote sensing technologies, can revolutionize data analysis (Janga et al., 2023). Artificial intelligence in remote sensing offers advantages such as improved data analysis capability, data quality, and accuracy, in this case, for mangrove classification and detection of cover changes. These methods have primarily been implemented in countries such as China, Australia, India, Thailand, Indonesia, Vietnam, among others (Janga et al., 2023; Pham et al., 2019; Velázquez-Salazar et al., 2021; Xia et al., 2020).

Although remote sensing data and artificial intelligence have been

studied extensively in other countries and methods have been shown to be effective, there is a paucity of implementation in countries such as Mexico, even though it is the fifth largest mangrove country in the world (Zu-Ermagassen and Spalding, 2021). Moreover, although it has been demonstrated that there are differences between species with respect to their spectral responses (Rahmandhana et al., 2022; Valderrama-Landeros et al., 2018), there is still a paucity of studies that explore the lack of specificity to create models that can identify distribution at the species level. This offers two potential avenues for further research: the opportunity to create models for unstudied regions and for species-level identification. Previous studies have investigated the potential of various deep learning artificial intelligence models, including neural networks, and have obtained excellent results (Ghorbanian et al., 2022; Kong et al., 2019; Wang et al., 2008). Therefore, the present study took the opportunity to implement the multilayer perceptron (MLP) model, a neuronal network interconnected in a layered structure. This model belongs to the category of machine learning and deep learning multi-class models, due to its ability to solve a range of problems, including those related to pixel-level classification and semantic segmentation, through its simple or complex configuration. The model comprises three layers: the input layer, which contains the data to be classified. Subsequently, the hidden layer that receives the data from the previous layer, which may be the input layer or another hidden layer. It then processes and transforms the information. Finally, the output layer, which provides the outcome of the network process (Russel and Norving, 2010).

Since the beginning of the century until the present, these models have been used in some environmental issues that also use remote sensing data (Gross et al., 1999; Keiner and Yan, 1998; Kimes et al., 1996; Trujillo-Acatitla et al., 2022). In some countries, models for monitoring mangroves have been developed using different methods of artificial neuronal networks (ANN) using data from passive and active sensors (Rahmandhana et al., 2022; Wang et al., 2018; Zulfa et al., 2021). Multilayer perceptron is one of the most widely used ANN methods currently in use, as it is able to extract complex features that help improve classification compared to traditional methods (Chollet, 2021; Goodfellow et al., 2017). The satellite data allows the identification of unique patterns in pixel values across multiple spectral bands representing different vegetation types (Chopade et al., 2023; Ghorbanian et al., 2022). In the case of the present study, we sought to achieve a more specific level, at the species level.

Although these methods have been implemented in some parts of the world, further exploration is needed in Latin America and Mexico. It is important to consider development at different scales, as the species present in each region and their spectral responses differ. Integrating technology, remote sensing, fieldwork, and artificial intelligence can enhance ecological monitoring for preserving ecosystems such as mangroves. This article presents an innovative methodology that integrates unmanned aerial vehicle (UAV) and satellite data with artificial intelligence models to identify and monitor mangroves. Furthermore, this methodology is expected to offer a more efficient and effective monitoring solution, leading to reduced costs in terms of economic, human, and time resources. Additionally, to the fact that this approach can be extended to other types of vegetation, making it a valuable tool to address the Sustainable Development Goals of the 2030 Agenda. Specifically, it supports Goal 15, which aims to sustainably manage forests, that entails the protection and restoration of terrestrial ecosystems, the combating of desertification, the halting and reversal of land degradation, and the halting of biodiversity loss.

2. Material and methods

Fig. 1 shows the method employed to acquire and process data for the current study. The data was gathered from multiple sources, including drone and satellite sensors, to create binary and multiclass mangrove databases for deep-learning models to monitor mangroves.

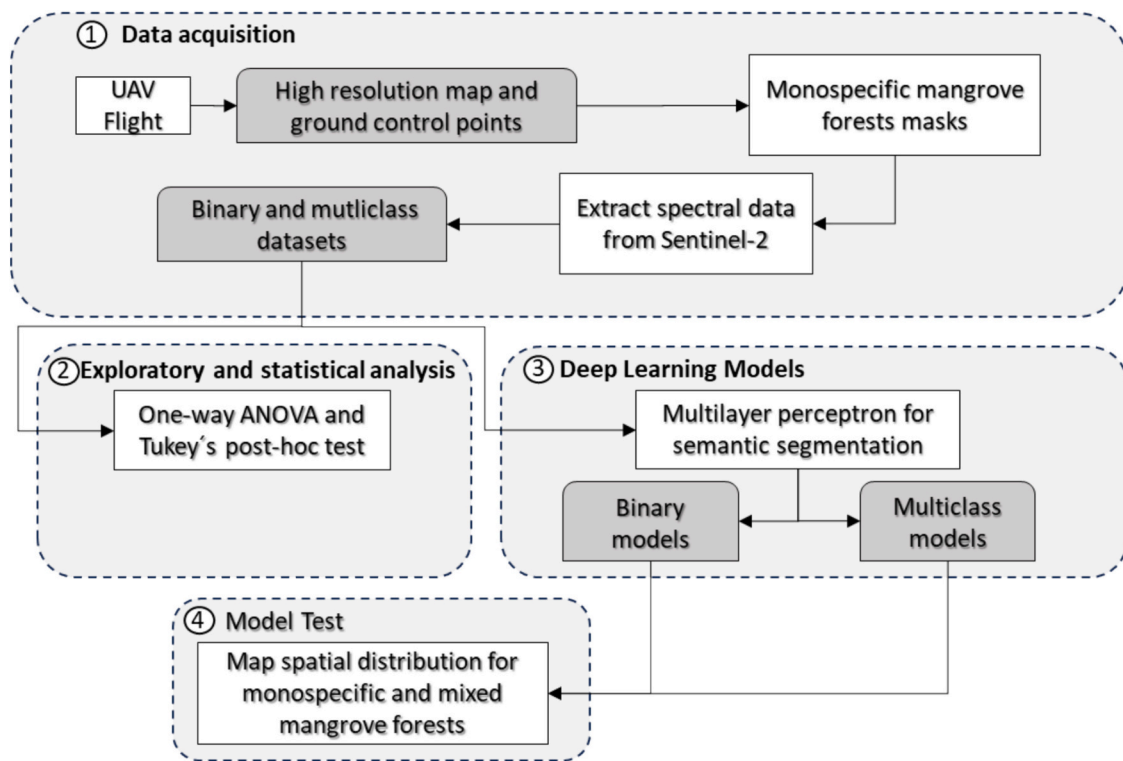


Fig. 1. Diagram of the method for the data acquisition with UAV to obtain a high-resolution map and used it as a base for creating a mangrove forest species mask, to extract then the spectral data from the satellite Sentinel-2. After this, an exploratory and statistical analysis was done with a one-way ANOVA and Tukey's post-hoc test. Next, deep learning models of multilayer perceptron models were trained and validated. Finally, a model test was implemented to create a map of spatial distribution for monospecific and mixed mangrove forests to evaluate the performance of the models.

2.1. Data acquisition

To achieve a more precise estimation of the distribution of mangroves, field data of the actual distribution of each species was collected in La Mancha, Veracruz and Arroyo Moreno, Veracruz Fig. 2. The site of La Mancha (19° 35' 12" N, 96° 23' 09" W), Veracruz, Gulf of Mexico with 9050 km² is a priority mangrove conservation site that has been a study site for various researchers, producing maps and environmental data. It was chosen as a pilot site due to the presence of well-defined monospecific forests and easy accessibility to obtain control points for vegetation verification for high-quality data.

A DJI Mavic Mini 2 UAV was used to conduct a zenithal drone shot of the area at an altitude of 100 m in July 2021. A total of 5435 photos were processed using the photogrammetry application Agisoft Metashape Professional (Version 1.8.5) to obtain an orthophoto of the mangrove forest area with the highest possible precision. Additionally, walking and boat trips were conducted to verify the distribution of mangrove species at 100 ground control points with a GPSMAP527. Later, polygons of updated masks of monospecific of red (*Rhizophora mangle*), black (*Avicennia germinans*) and white (*Laguncularia racemosa*) mangrove forest; and mixed mangroves with flooded freshwater forested wetlands were created using QGIS version 3.22.14, concerning the orthophoto and field verification points.

Subsequently, a total of 147 cloud-free Sentinel-2 L1C image datasets of the period 2015–2021 were obtained from the Copernicus Browser of the European Union's Space programme and digital numbers were converted into surface reflectance values. Also, the red-edge bands (Band-5 0.705 µm, Band-6 0.740 µm, Band-7 0.783 µm and Band-8 A 0.865 µm) and shortwave infrared bands (Band-11 1.610 µm and Band-12 2.190 µm) were resampling using the bilinear method with the rasterio package in Python 3.8.13 to achieve a spatial resolution of 10 m, such as the red, green, blue, and near-infrared bands.

The created polygons of the distribution of the mangrove forest were

used to extract the spectral information from the pixel values for each band. The dataset was then normalized and standardized using the standard scaler based on standard normally distributed data from Scikit learn 1.2.1 in Python 3.8.13. Two class-balanced databases were created using the TOA pixel values from Sentinel-2. The first one was a binary database containing 60,000 pixel data according to what is mangrove, where the mask of the three species were considered as one, and non-mangrove; the second was a multiclass database containing 40,000 pixel data with four classes, created with each mask of the three species (*R. mangle*, *L. racemosa*, and *A. germinans*) and non-mangrove areas. The non-mangrove class was selected randomly from data belonging to different classes including rainforest, crops, water, urban, dunes, and other kind of wetlands.

2.2. Exploratory and statistical analysis

A simple variance analysis (One-Way ANOVA) was performed on each band to evaluate whether there was a significant difference between species. A Tukey's post-hoc test was then conducted with a confidence interval of 95 % (α of 0.05) to identify which species were different. The bioinfokit v1 package in Python 3.8.13 was used for this process (Kutner et al., 2004).

2.3. Multilayer perceptron for semantic segmentation

The multilayer perceptron is the best known and most widely used type of neural network, whose architecture consists of multiple layers of interconnected neurons that could be used for a sub-pixel classification. It has non-linear activation functions, such as RELU, which allow it to recognize complex patterns in data. It also has optimization algorithms, such as ADAM, that improve model learning (Goodfellow et al., 2017).

In this study, the initial model consisted of a single input and one output layer, with the sigmoid as the activation function, so that it could

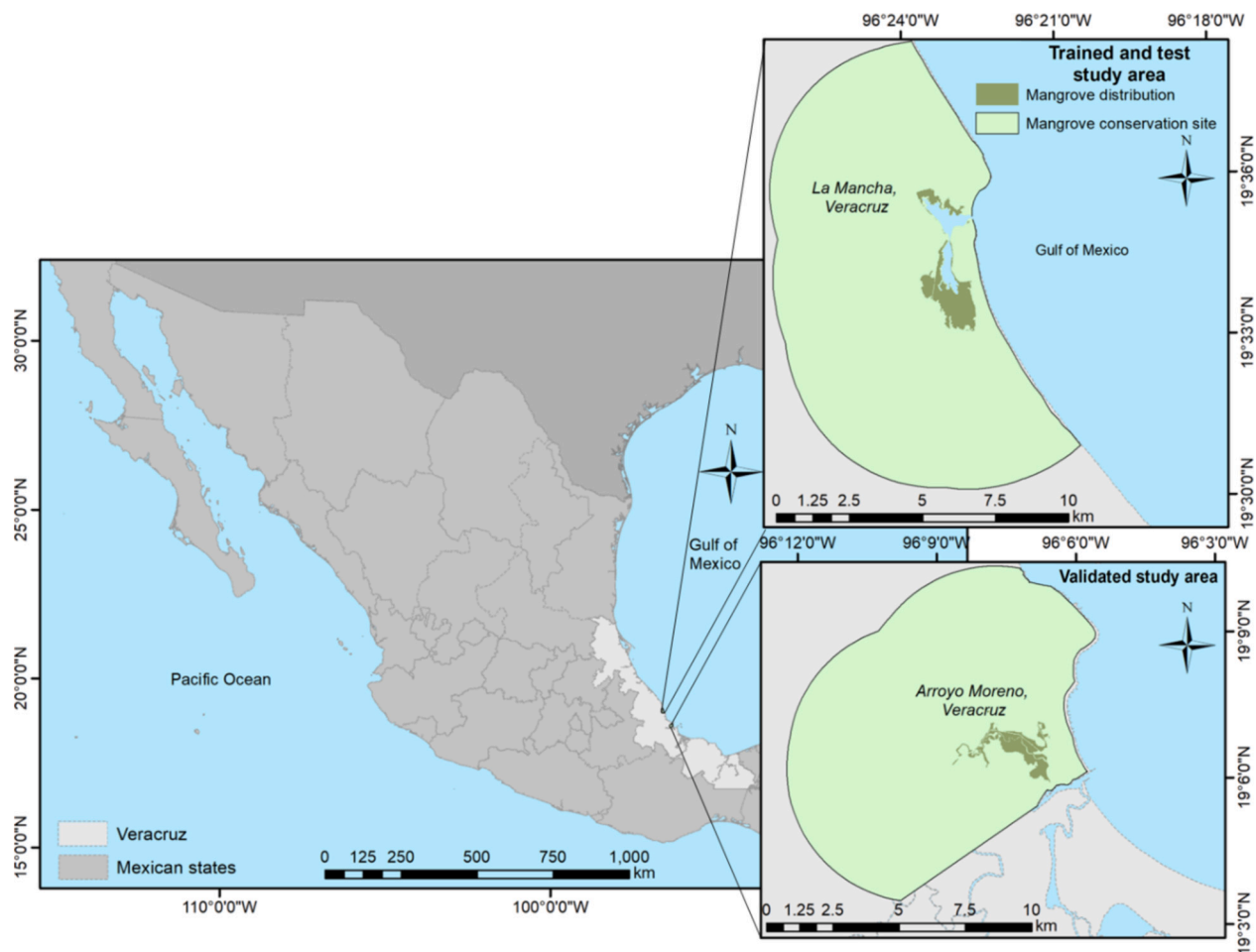


Fig. 2. Study area of La Mancha, Veracruz, Mexico.

be interpreted as a logistic regression. Subsequently, various architectures were explored to determine the optimal configuration for the MLP model, ranging from one to five hidden layers, each one containing multiple neurons or neurons (5, 10, 20, 50, 100, 150, 300, 500). In all of them the relu activation function in the hidden layer was implemented, and ADAM was employed as the optimizer.

The database used for the models includes reflectance data from the RGB bands (Band-2 0.490 μm , Band-3 0.560 μm , Band-4 0.665 μm), Vegetation Red Edge bands (Band-5 0.705 μm , Band-6 0.740 μm , Band-7 0.783 μm), Near-Infrared bands (Band-8 0.842 μm and Band-8 A 0.865 μm) and Short-wave Infrared bands (Band-11 1.610 μm and Band-12 2.190 μm). Then the dataset was split into training and test sets using the train test split module of scikit-learn 1.2.1, with 70 % for training and 30 % for test.

For the binary models, the binary crossentropy was used as the loss function and the sigmoid activation function in the output layer was used, this one produces values ranging from 0 to 1, where 0 represents the non-mangrove, and 1 represents mangrove. For the multiclass segmentation, multiclass cross-entropy was used as the loss function, and the softmax activation function in the output layer was used. The output layer of the multiclass models produces a probability for each class; the one with the highest probability is assigned to the corresponding class, 0 for non-mangrove, 1 for red mangrove (*R. mangle*), 2 for black mangrove (*A. germinans*), 3 for white mangrove (*L. racemosa*).

Finally, a recent image of Sentinel-2 was used to assess the models' performance test in La Mancha, where the acquisition data was taken and from Arroyo Moreno (19° 00' 59"N, 96° 03', 43"W), in Veracruz.

3. Results

The results of this study are presented below. Fig. 3 shows the high-resolution map obtained with a resolution of 3.5 cm and the masks created to represent the ground truth of mangrove species.

3.1. Exploratory and statistical analysis

Fig. 4 shows the boxplots for each band and species (*R. mangle*, *A. germinans*, and *L. racemosa*). The One-Way ANOVA and Tukey *post-hoc* test reveal a significant statistical difference between the species in all the cases ($p\text{-value} < 0.05$). This indicates that there are variations in the spectral responses for each species, enabling distinctions to be made between monospecific forests.

3.2. Multilayer perceptron models

The binary MLP accuracies are shown in Fig. 5, where all of them exceed the 0.990. The lowest values were obtained in the logistic model, with a training accuracy of 0.9916 and a test accuracy of 0.9910. The MLP models with the highest scores were those of a hidden layer with 300 neurons, achieving a training accuracy of 0.9995 and a test accuracy of 0.9988; two hidden layers with 100 neurons achieved a training accuracy of 0.9994 and a test accuracy of 0.9987; three hidden layers with 50 neurons, achieved a training accuracy of 0.9996 and a test accuracy of 0.9987; four hidden layers with 150 and 300 neurons achieved a training accuracy of 0.9993 and a test accuracy of 0.9986; and finally, five hidden layers with 50 and 300 neurons achieved a training accuracy

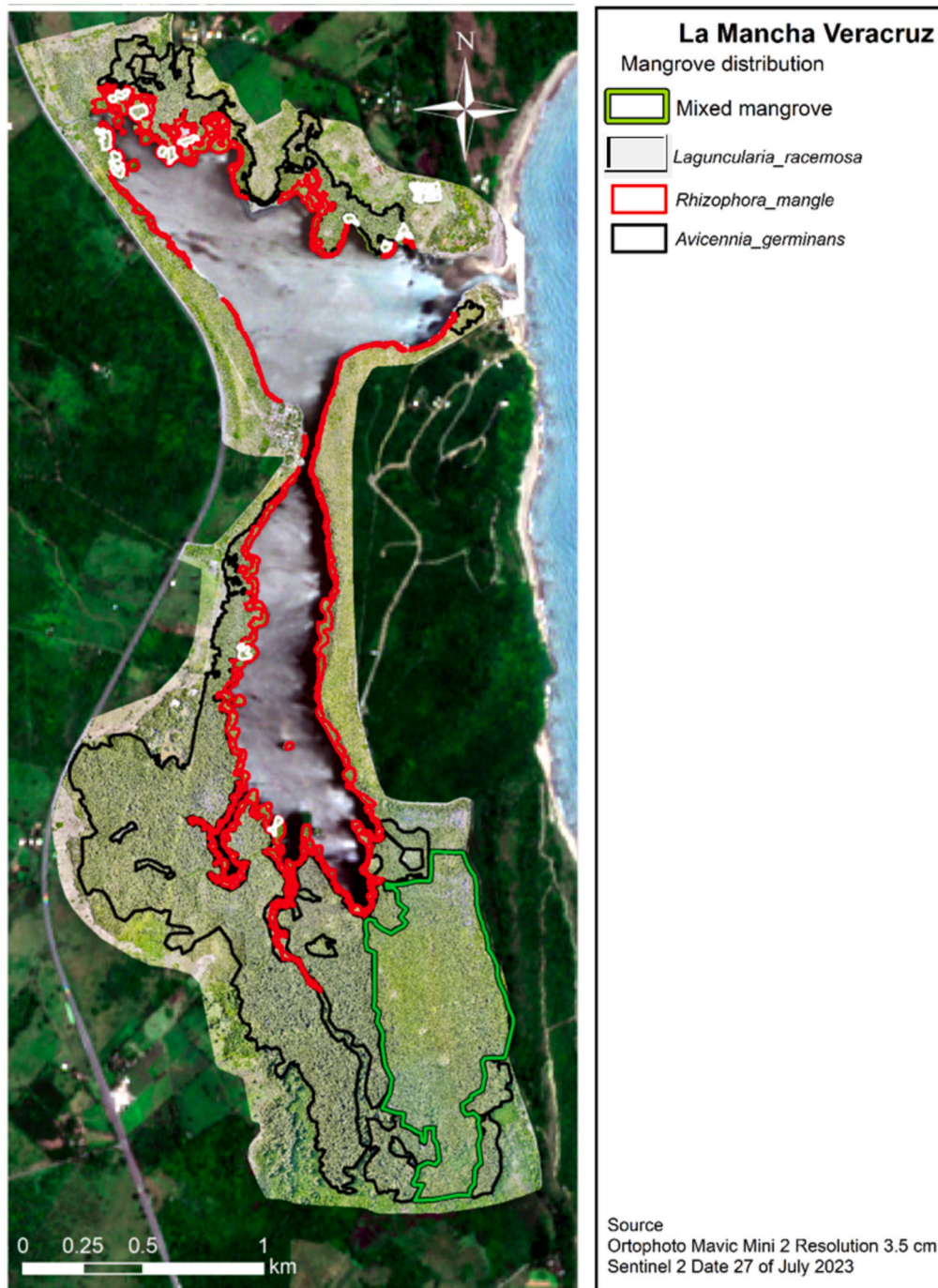


Fig. 3. Ortophoto and mangrove masks in La Mancha, Veracruz.

of 0.9995 and a test accuracy of 0.9988. Upon comparison, the model comprising three hidden layers and 50 neurons demonstrated the most optimal scores across all.

The multiclass MLP accuracies are shown in Fig. 6, all the models obtained accuracies over 0.90, except for the logistic model which obtained the lowest values, a training accuracy of 0.7670 and a test accuracy of 0.7684. The model with one hidden layer and 500 neurons achieved a training accuracy of 0.9110 and a test accuracy of 0.9065; the model with two hidden layers and 300 neurons achieved a training accuracy of 0.9575 and a test accuracy of 0.9436; the model with three hidden layers and 500 neurons achieved a training accuracy of 0.9863 and a test accuracy of 0.9654, which had the highest accuracies; the model with four hidden layers and 300 neurons achieved a training

accuracy of 0.9795 and a test accuracy of 0.9602; and finally, the model with five hidden layers and 300 neurons achieved a training accuracy of 0.9873 and a test accuracy of 0.9684.

3.3. Models test

Fig. 7 shows the performance test of the spatial distribution of mangrove forest in binary models with the highest accuracies to validate them. The model of four hidden layers with 300 neurons was the most reliable, while the logistic model was the least reliable. It can be noted that, despite the similar accuracies, the performance in terms of spatial distribution markedly improves as the number of layers increases. When models with the same accuracy and number of layers are compared, the

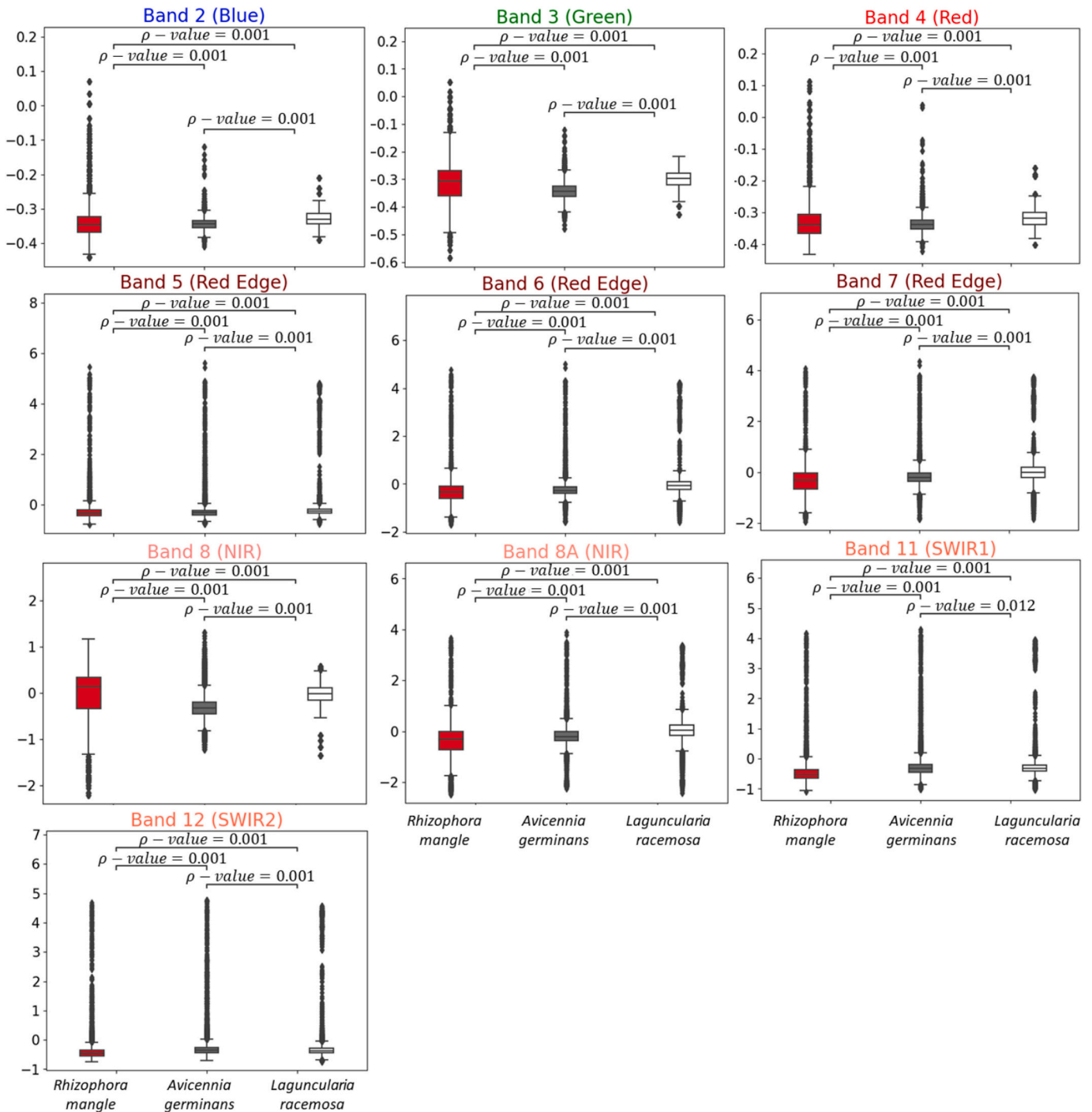


Fig. 4. Boxplot for each of the species (*R. mangle*, *A. germinans*, and *L. racemosa*) by band. In all the comparisons, there is a significant statistical difference between the species ($p < 0.05$).

distribution is underestimated as the number of neurons increases. This is evident in the four-layer models with 150 and 300 neurons, as well as the five-layer models with 50 and 300 neurons. The models with the highest accuracies and comprising three or more layers had better performance when tested.

The performance test of the spatial distribution of mangrove species in multiclass models with the highest accuracies to validate them are presented in Fig. 8, which contrasts with the previous models. However, once again, the model of four hidden layers with 300 neurons was the most reliable, while the logistic model was the least reliable. In general, *R. mangle* colored in red presented difficulties in differentiating itself

from other classes, such as water, mixed forest mangrove and agricultural areas. Conversely, *L. racemosa* colored in grey exhibited an over-estimation for the logistic model. In contrast, the performance of *A. germinans* colored in black was superior compared to the rest of the species.

A detailed examination of the multiclass models revealed that, despite having a lower score, but exceeding 0.95, some models demonstrated more precise performance across all the species spatial distributions Fig. 9. The model of four hidden layers with 100 neurons achieved a training score of 0.9662 and a test score of 0.9528, whereas the one hidden layer with 150 neurons attained a training score of

Multiperception models for mangrove

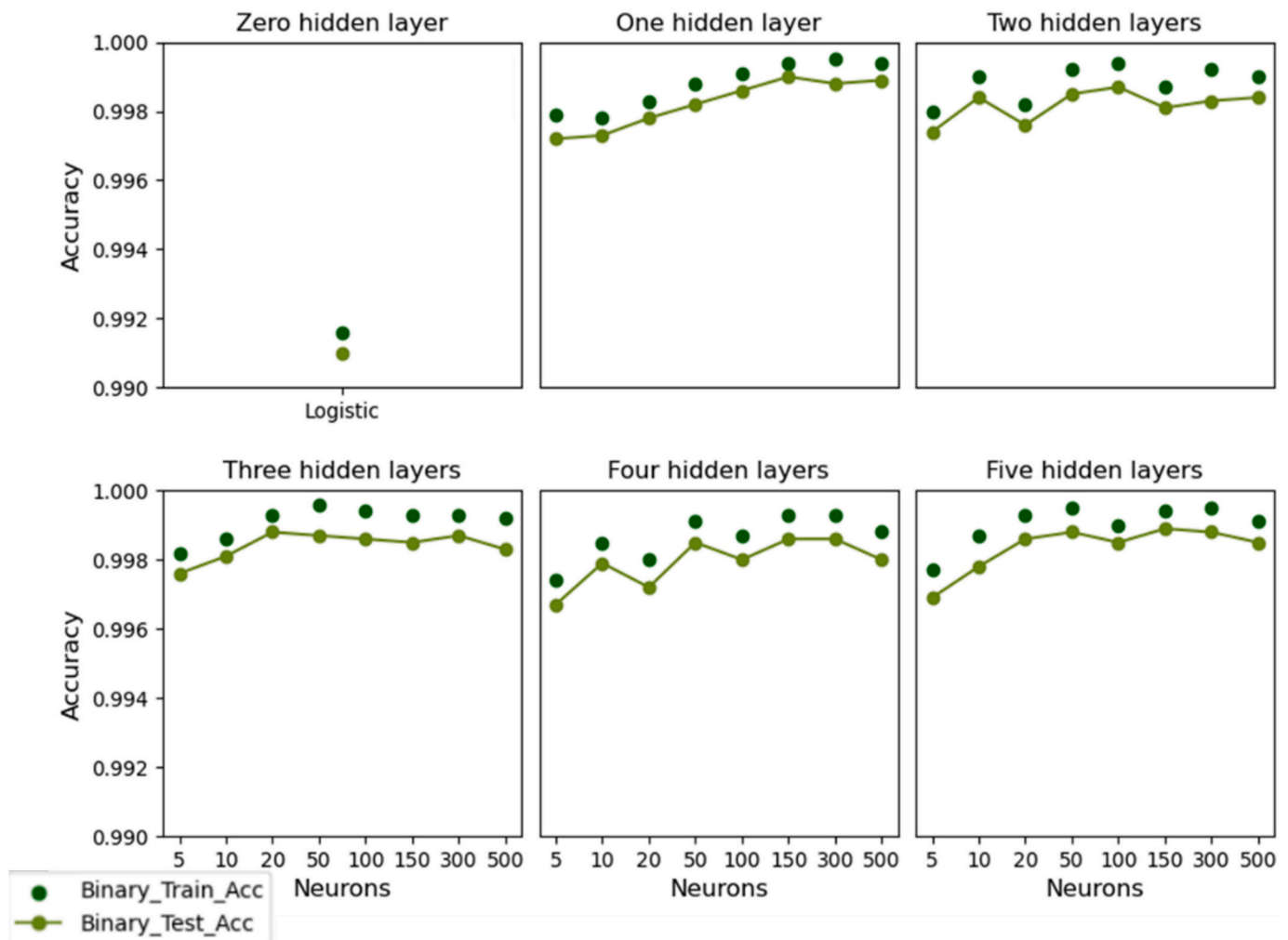


Fig. 5. Accuracy of training and test binary MLP models.

0.9775 and a test score of 0.9607. Conversely, the model of five layers with 50 neurons achieved a training score of 0.9550 and a test score of 0.9454, while the one with 150 neurons attained a training score of 0.9810 and a test score of 0.9632. Of all models, the most reliable was the one of five hidden layers with 50 neurons.

Finally, to assess the scalability and generalizability of the models, tests were carried out in other sites where mangrove distribution is known. A performance test was therefore conducted in Arroyo Moreno, Veracruz (Fig. 10), where the distribution of mangrove species has been previously described in other studies (López-Portillo et al., 2002) and where control points were taken for verification in the present study. The accuracy of the mangrove distribution models was remarkable but it should be mentioned that there is an overestimation with *Rhizophora mangle*. Nevertheless, the models demonstrated a robust approximation, with the four-layer models exhibiting superior performance compared to the five-layer models. However, taking into account the overestimates, again the most reliable of all models was the five hidden layer model with 50 neurons.

4. Discussion

Over time, several studies have used multispectral data from different satellites for vegetation monitoring, including the development of mangrove-specific indices such as Mangrove Vegetation Index (MVI)

and Combined Mangrove Recognition Index (CMRI) (Chen et al., 2023; Roy et al., 2019; Tran et al., 2022). However, the results of the present study have shown that by exploiting the potential of the technologies together, given that there are spectral differences in the same bands between species, multispectral data can be used directly to identify mangrove distribution without the need for additional processing, such as the use of vegetation indices.

Research on differences in spectral responses for mangroves has mainly been conducted in countries such as China, India, and the United States. However, there is a paucity of comprehensive studies on Latin American species in this field (Arun Prasad and Gnanappazham, 2014; Blasco et al., 1998; Rahmandhana et al., 2022; Zulfa et al., 2021). Avina-Hernández et al. (2023) is one of the few mangrove studies of Latin America that employ artificial intelligence, utilizing drone images within a spectral range of RGB and GRN (Green, Red and Near-infrared) with a maximum accuracy of 0.77, in comparison to the present studies, which exhibit a higher accuracy of over 0.90. Therefore, further research is necessary in Latin America, an area with high biodiversity and extensive mangrove areas, to better understand the region's spectral responses to mangroves and its potential for the development of monitoring methods with artificial intelligence (Rahmandhana et al., 2022). Even though the mangrove forest in Mexico covers 60 % of the coastline there are few machine learning models developed for ecological applications and limited information on the spectral features of mangrove

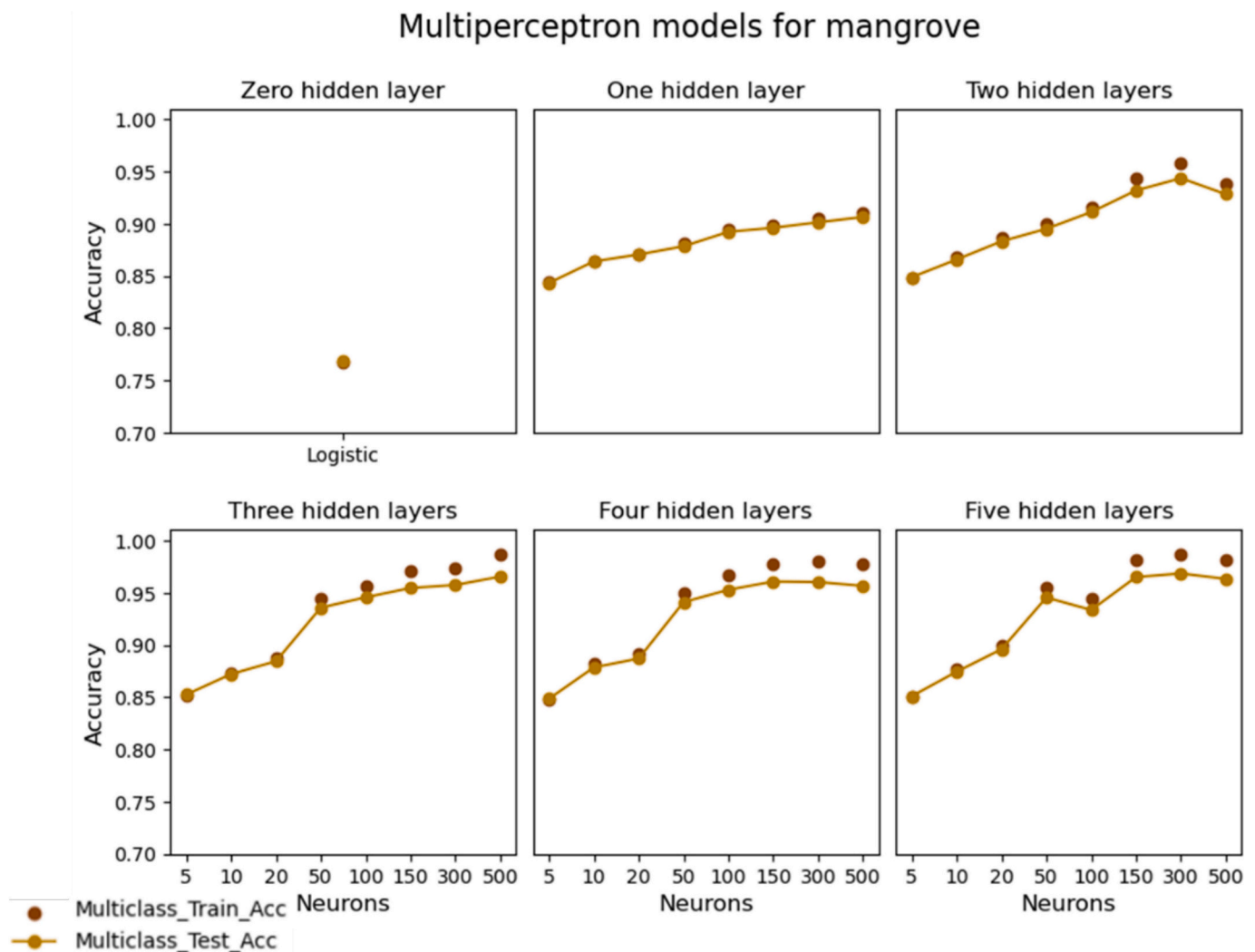


Fig. 6. Accuracy of training and test multiclass MLP models.

species in Mexico (Aviña-Hernández et al., 2023).

However, it is important to assess the suitability of artificial intelligence models, focusing on species distribution at regional scales. In other countries, models of artificial intelligence have been well-developed, with species of mangroves distributed in different regions using data from multiple sensors (Rahmandhana et al., 2022; Wang et al., 2018; Xia et al., 2020). Nevertheless, the development of these methodologies requires fundamental consideration of the conditions, diversity and configuration of the ecosystems. According to some studies, mangroves appearances and spectral features vary depending on the region and species, making it challenging to generalize the models (Aji et al., 2023; Guo et al., 2021).

It is possible that, given this situation, there are not many studies that address the identification of vegetation at the species level. For this reason, an advantage from the study area was taken since the presence of monospecific mangrove forests of the three predominant species of mangrove in Mexico. This offered an opportunity to obtain pure multispectral data for each species and to deepen our understanding of their specific spectral responses, in contrast to other locations where there is a mingling of patches with different mangrove dominance and compositions. In addition, many mangrove species are widely distributed throughout the world. The red mangrove (*R. mangle*) is found in 34 countries, the black mangrove (*A. germinans*) in 38 countries and the white mangrove (*L. racemosa*) in 33 countries. Their distribution covers the continents of America and Africa, as well as island areas (Plants of

the World Online, 2024). Consequently, the artificial intelligence models developed in this study are applicable at a global level, extending beyond the scope of Mexico (Lacerda, 2002).

This was used as an opportunity to develop two distinct models with the objective of creating a multi-purpose tool. The first model differentiates between mangrove and non-mangrove areas, regardless of the species, in order to be employed in ecosystems where mangroves co-occur with other vegetation. The second model was focused on monospecific forest mangrove to facilitate conservation decision-making or regeneration needs for specific species. In both cases, the models are applicable to Mexico and different regions of the world, where there are only a few studies employing similar methodologies (Aviña-Hernández et al., 2023; Cissell et al., 2017). Although studies in countries such as Iran and Vietnam have developed similar methods for mangrove identification (Ghorbanian et al., 2022; Xia et al., 2020; Zu-Ermagassen and Spalding, 2021), these have not been developed at the species level and cannot be generalized to other countries with different species compositions.

The results demonstrated differences in performance between binary and multi-class models, which can be attributed to the inherent variability of the multispectral response, as previously pointed out by several authors (Arun Prasad and Gnanappazham, 2014; Zulfa et al., 2021). Also, the results indicated the necessity of assessing the efficacy of the models, as the highest accuracies did not necessarily yield the most accurate spatial distributions when testing the models. As

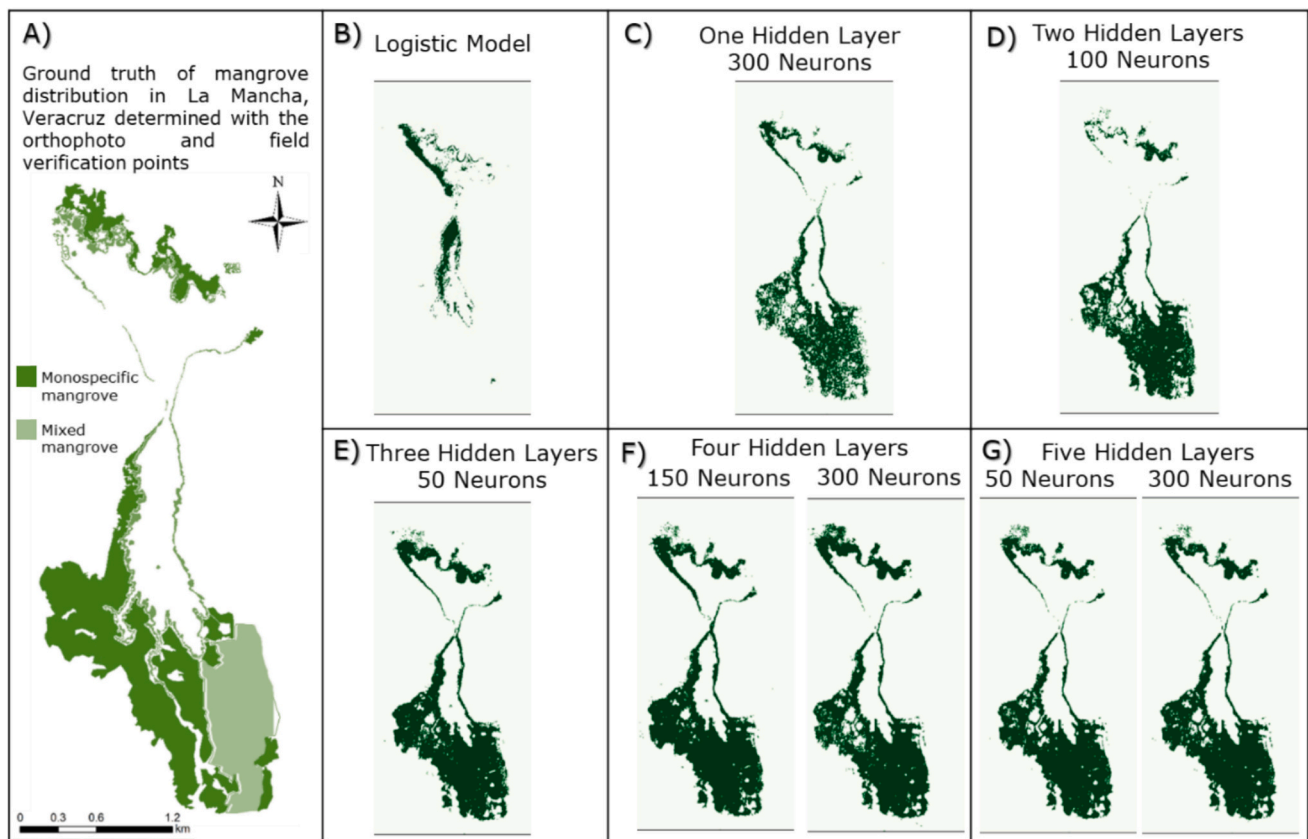


Fig. 7. Binary model performance of the mangrove spatial distribution of the most accurate models at La Mancha, Veracruz. Mangrove area is colored in green. A) Ground truth from the mangrove distribution in La Mancha, Veracruz determined with the orthophoto and field verification points, the mixed mangrove represented in a lighter shade; B) Logistic Model; C) Model of One Hidden Layer with 300 neurons; D) Model of Two Hidden Layers with 100 neurons; E) Model of Three Hidden Layers with 50 neurons; F) Models of Four Hidden Layers 150 and 300 neurons; G) Models of Five Hidden Layers with 50 and 300 neurons. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Kumaraperumal et al. (2022) points out, is it important to consider the nature of the datasets, the complexity and structure of the objects of study, such as mangroves. This given the importance of taking into account that the performance of a model is not necessarily guaranteed by a high accuracy, given the inevitable presence of biases that can affect its performance with new data, and needs to be addressed as mentioned by Chowdhury (2024). In this case, some models, although obtaining high accuracy, showed errors in their performance and confusion with water. This could be due to the reflectance of mangrove, which has a higher moisture content than other vegetation types, making it difficult to distinguish in the analysis (Yang et al., 2022).

Furthermore, in contrast to the typical community-level approach to mangrove monitoring, our models demonstrated the ability to accurately predict the distribution of monospecific and mixed mangroves, a scope that had not previously been attempted. The mixed mangrove described in this study has a complex composition with the presence of mangrove and low deciduous forest. According to multiclass models, the mixed mangrove is dominated by black mangrove (*A. germinans*) in the area closest to the lagoon, followed by the presence of white mangrove (*L. racemosa*) and, further south, red mangrove (*R. mangle*). Some studies mention the dominance of black mangrove (*A. germinans*) and white mangrove (*L. racemosa*) in this area, with the presence of flooded freshwater forested wetlands (López-Portillo et al., 2002). Others report the presence of secondary mangrove vegetation, also with a predominance of mainly black (*A. germinans*) and white (*L. racemosa*) mangrove (Environmental-Consultancy, 2022; Hernández-Trejo et al., 2006). Red mangrove (*R. mangle*) dominance is also reported in some permanently flooded areas in the region (Environmental-Consultancy, 2022). This is consistent with the models and demonstrates the potential of multi-class

models to identify mangrove species dominance, despite having mixed pixels with other vegetation types.

In the multiclass models, the variability of the data increases, which makes them more sensitive in comparison to the binary models. In this sense, it is important to highlight that MLP model approximations can vary according to their architecture, and the comparison between the results in Figs. 8 and 9 shows that a model with fewer neurons can offer significant advantages in terms of simplicity, generalisation and effectiveness in finding relevant patterns with more efficient performance in identifying monospecific forests, thus avoiding overfitting (Hagan et al., 2014). This is therefore a demonstration of the potential of multi-class artificial intelligence models for the monitoring of species-specific distribution changes.

With regard to the artificial intelligence methods employed, several studies have shown that ANN can achieve similar accuracies using algorithms such as RetinaNet (Chen et al., 2023; Ghorbanian et al., 2022) when using multispectral data from Sentinel-2. Other research indicates that incorporating multiple indices in the input features can significantly enhance and improve the models. However, this study only employed the original multispectral data and obtained better accuracies, which indicates that obtaining the data in its purest form for model training is preferable, rather than relying on indices (Guo et al., 2021; Maung and Sasaki, 2021).

Although the National Commission for the Knowledge and Use of Biodiversity (CONABIO) in Mexico has implemented a mangrove monitoring program on a five-year cycle, a robust monitoring method for mangroves has not yet been fully developed. To date satellite images have been employed with a spatial resolution of 250 m, such as MODIS, others with a resolution of 30 m, such as Landsat, and with a resolution

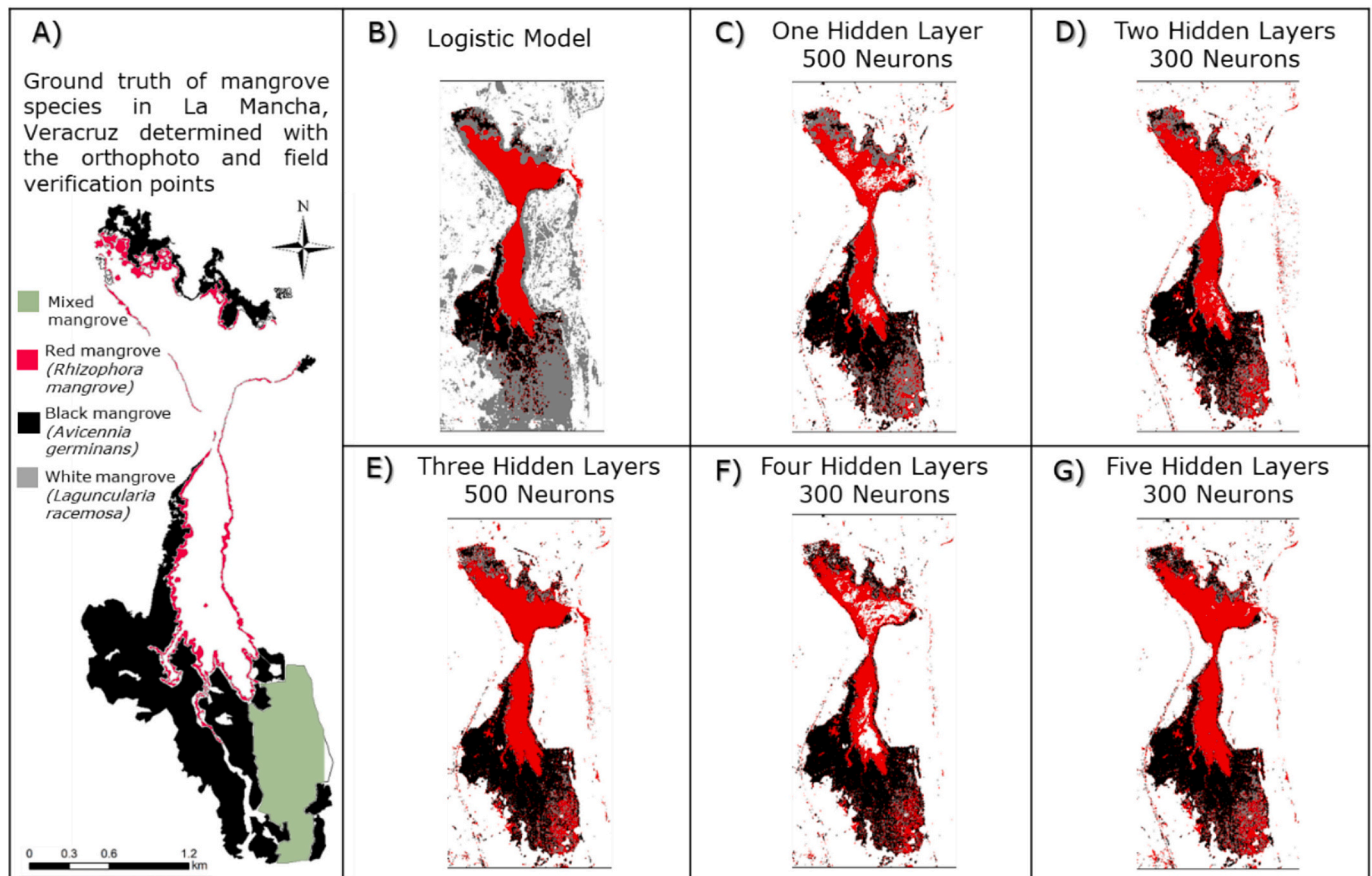


Fig. 8. Multiclass models performance of the highest accuracies models in La Mancha, Veracruz. Each colour corresponds to one of the species: the red mangrove (*R. mangle*) is colored in red, the black mangrove (*A. germinans*) is colored in black, and the white mangrove (*L. racemosa*) is colored in grey, everything that is not mangrove is colored in white and the mixed mangrove is colored in a lighter shade. A) Real mangrove distribution by species in La Mancha, Veracruz determined with the orthophoto and field verification points; B) Logistic Model; C) Model of One Hidden Layer with 500 neurons; D) Model of Two Hidden Layers with 300 neurons; E) Model of Three Hidden Layers with 500 neurons; F) Models of Four Hidden Layers with 300 neurons; G) Models of Five Hidden Layers with 300 neurons. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

of 10 m, such as SPOT (European Space Agency, 2014). This has implications with respect to the modifiable areal unit problem, which consists of inconsistency in the analysis of results when using data at different scales (Jelinski and Wu, 1996). This problem can result in a significant variation when estimating the extent of mangroves in Mexico, causing a very certain overestimation or underestimation of the mangrove distribution, as data obtained at different scales may not be comparable. Despite this, the CONABIO has asserted that the accuracy of the mangrove cartography at a scale of 1:50,000 was estimated to be 90.5 % (CONABIO, 2009) increasing to 95 % for more recent maps (Rodríguez-Zuñiga et al., 2013). The best models obtained in the present study achieved 99.8 % accuracy for binary models such as those described by CONABIO, and over 98 % of accuracy for the multiclass models. These models are therefore very useful for monitoring mangroves.

Consequently, the present study proposes a solution involving the use of open access satellites such as Sentinel-2 data, widely used for monitoring vegetation worldwide, with a fine resolution of 10 m. This is suitable for vegetation monitoring in comparison to the satellite SPOT data, which has a significant economic cost and limited benefits in terms of spectral and spatial resolution. Furthermore, the proposed solution offers a reduction in the human effort required for field sampling, as well as the potential to reduce the human risks from fieldwork to determine changes after a natural disaster. Also, this is coupled with the opportunity to be extrapolated to any part of the world by integrating enough species-specific data.

4.1. Considerations and future research

Previously, mangrove monitoring was conducted through field surveys, light aircraft or helicopter flights, and the use of costly satellite imagery. However, technological advances have enabled the use of drones for high-spatial-resolution mapping and open-access satellites, such as Sentinel-2, for the acquisition of a vast quantity of multispectral data. This has the dual benefit of reducing economic costs, human effort and improving data quality. Furthermore, artificial intelligence models have demonstrated considerable potential for ecological applications in the monitoring of ecosystems such as mangroves.

However, several factors warrant consideration. It has been demonstrated that there is variation in reflectance between species. Consequently, the pixel information in a satellite image may comprise a mixed composition of reflectances. It is therefore essential to consider this factor when interpreting the results of artificial intelligence models. This manifested as an erroneous classification of the red mangrove (*R. mangle*) as water in some multi-class models.

Similarly, it is crucial to consider the composition of the ecosystems and the degree of generalisation inherent to the models. In the present study, mixed mangrove areas were considered; however, it should be noted that mangroves can vary in their reflectance depending on the presence, diversity and dominance of species. The present study has demonstrated the potential for mapping at a more specific level with these technologies. However, as knowledge of this is further developed and integrated, the method can be improved to other ecosystems and

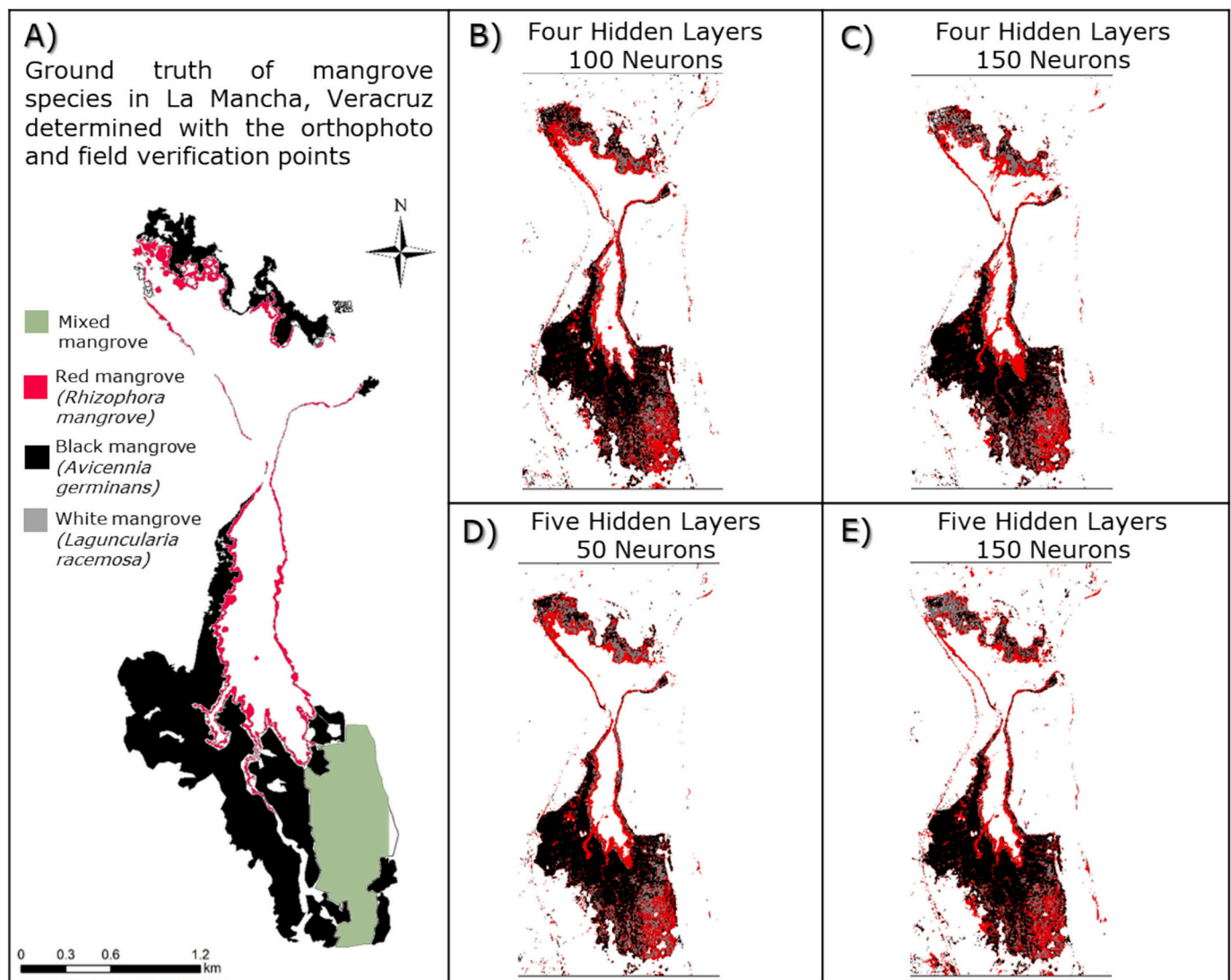


Fig. 9. Best multiclass models' performance from La Mancha, Veracruz. Each colour corresponds to one of the species: the red mangrove (*R. mangle*) is colored in red, the black mangrove (*A. germinans*) is colored in black, and the white mangrove (*L. racemosa*) is colored in grey one, everything that is not mangrove is colored in white and the mixed mangrove is colored in a lighter shade. A) Real mangrove distribution by species in La Mancha, Veracruz determined with the orthophoto and field verification points; B) Model of Four Hidden Layer with 100 neurons; C) Model of Four Hidden Layer with 150 neurons; D) Model of Five Hidden Layers with 50 neurons; E) Model of Five Hidden Layers with 150 neurons. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

extended to larger and larger regions.

Finally, as a future work, it should be considered that the MLP models, which have shown certain errors in the detection of mangroves, could later opt for more advanced structures. This implies the use of segmentation models that incorporate contextual information in addition to spectral response values, based on convolutional neural networks (CNN) such as UNET.

5. Conclusions

The acquisition of data from multiple sources confers several advantages, which expand the possibilities beyond traditional monitoring methods, which are typically carried out using *in situ* measurements in the field to verify the ground truth, which incurs in high economic, human, and time costs. The use of UAVs offers the potential to obtain information on the distribution of mangroves in areas that are difficult to access. Conversely, multispectral data from optical satellites, such as Sentinel-2, is freely available and provides consistent information over time. Finally, the application of artificial intelligence algorithms as a

novel tool has proven useful for differentiation vegetation down to the species level, which opens up new possibilities for mangrove monitoring, where the development and implementation of these new technologies is needed.

This methodology may be employed as an alternative to monitoring monospecific and mixed mangrove forests, thereby affording the opportunity to identify areas of ecological vulnerability based on the condition of the mangroves. The objective of this study is to facilitate the monitoring of mangroves in Mexico and different regions of the world, thereby human risks from fieldwork through the implementation of remote monitoring. Furthermore, it is a cost-effective solution with high efficacy and efficiency, which will facilitate the implementation of long-term government programs aimed at achieving objectives such as mangrove mapping, conservation, and the enhancement of regeneration efficiency in the context of climate change and coastal vulnerability.

Description of author's responsibilities

Erandi Monterrubio-Martínez: Research, field data acquisition,

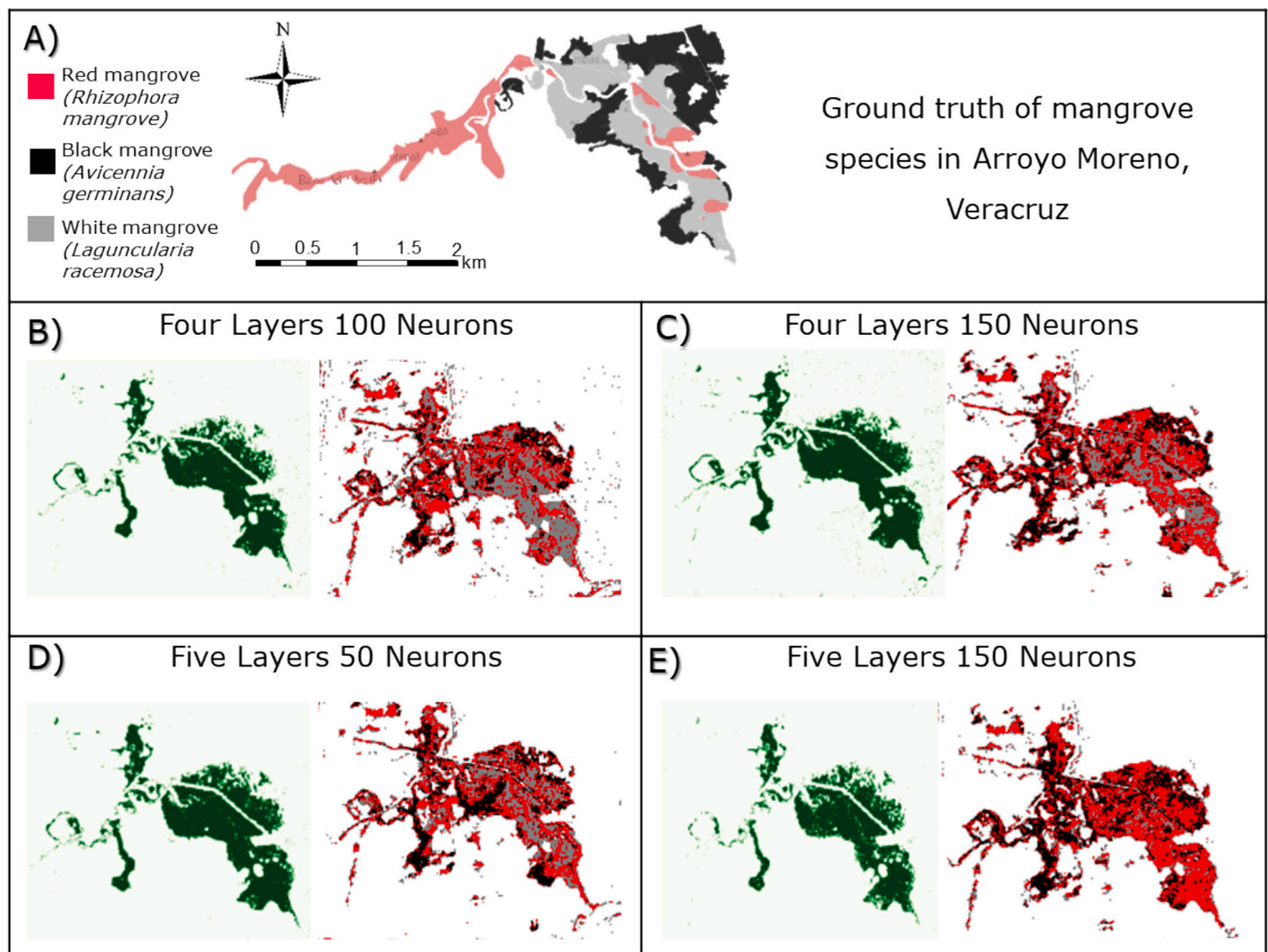


Fig. 10. Best binary and multiclass models' performance from Arroyo Moreno, Veracruz. Each colour corresponds to one of the species: the red mangrove (*R. mangle*) is colored in red, the black mangrove (*A. germinans*) is colored in black, and the white mangrove (*L. racemosa*) is colored in grey, everything that is not mangrove is colored in white. A) Real mangrove distribution by species in Arroyo Moreno (López-Portillo and Ezcurra, 2002); B) Model of Four Hidden Layer with 100 neurons; C) Model of Four Hidden Layer with 150 neurons; D) Model of Five Hidden Layers with 50 neurons; E) Model of Five Hidden Layers with 150 neurons. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

creation of image dataset, data curation, method, formal analysis, implementation of Deep Learning Models, original draft writing – review and editing. Rubicel Trujillo-Acatitla: Research, field data acquisition, implementation of Deep Learning Models, data curation, visualization. José Tuxpan-Vargas: Research, conceptualization, validation, writing, revision and editing, project administration, visualization, supervision. Patricia Moreno-Casasola: Research, conceptualization, validation, writing, revision and editing, project administration, visualization, supervision.

Declaration of competing interest

The authors declare no competing monetary interests or personal relationships to influence the work presented here.

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Data availability

Data is open access able in <https://knb.ecoinformatics.org/view/doi:10.5063/F1FN14P1> and code will be made available upon request.

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